

Assignment 4: Data Wrangling (Spring 2026)

Lex Clay

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Wrangling. Do not use any AI tools in completing this assignment.

Directions

1. Rename this file `<FirstLast>_A04_DataWrangling.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. **Ensure that code in code chunks is tidy and does not extend off the page in the PDF.**
7. **Push your completed RMD to your GitHub account**

Set up your session

- 1a. Load the `tidyverse` and `here` packages into your session.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## Warning: package 'here' was built under R version 4.5.2
```

```
## here() starts at C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026
```

```
library(dplyr)
```

1b. Check your working directory.

```
getwd()
```

```
## [1] "C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026"
```

```
here()
```

```
## [1] "C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026"
```

1c. Read in all four raw data files associated with the EPA Air dataset, being sure to set string columns to be read in as factors. See the README file for the EPA air datasets for more information (especially if you have not worked with air quality data previously).

2. Add the appropriate code to reveal the dimensions of the four datasets.

```
#1a
```

```
#1b
```

```
#1c
```

```
EPAAirList <- list(  
  EPAAirOne <- read.csv(file=here('./Data/Raw/EPAAir_03_NC2018_raw.csv'),stringsAsFactors=TRUE),  
  EPAAirTwo <- read.csv(file=here('./Data/Raw/EPAAir_03_NC2019_raw.csv'),stringsAsFactors=TRUE),  
  EPAAirThree <- read.csv(file=here('./Data/Raw/EPAAir_PM25_NC2018_raw.csv'),stringsAsFactors=TRUE),  
  EPAAirFour <- read.csv(file=here('./Data/Raw/EPAAir_PM25_NC2019_raw.csv'),stringsAsFactors=TRUE)  
)  
  
str(EPAAirList)
```

```
## List of 4  
## $ : 'data.frame': 9737 obs. of 20 variables:  
## ..$ Date : Factor w/ 364 levels "01/01/2018","01/02/2018",...: 60 61  
## ..$ Source : Factor w/ 1 level "AQS": 1 1 1 1 1 1 1 1 1 1 ...  
## ..$ Site.ID : int [1:9737] 370030005 370030005 370030005 370030005 370030005 370030005 370030005 370030005 370030005 370030005 ...  
## ..$ POC : int [1:9737] 1 1 1 1 1 1 1 1 1 1 ...  
## ..$ Daily.Max.8.hour.Ozone.Concentration: num [1:9737] 0.043 0.046 0.047 0.049 0.047 0.03 0.036 0.036 0.036 0.036 ...  
## ..$ UNITS : Factor w/ 1 level "ppm": 1 1 1 1 1 1 1 1 1 1 ...  
## ..$ DAILY_AQI_VALUE : int [1:9737] 40 43 44 45 44 28 33 41 45 40 ...  
## ..$ Site.Name : Factor w/ 40 levels "", "Beaufort",...: 35 35 35 35 35 35 35 35 35 35 ...  
## ..$ DAILY_OBS_COUNT : int [1:9737] 17 17 17 17 17 17 17 17 17 17 ...  
## ..$ PERCENT_COMPLETE : num [1:9737] 100 100 100 100 100 100 100 100 100 100 ...  
## ..$ AQS_PARAMETER_CODE : int [1:9737] 44201 44201 44201 44201 44201 44201 44201 44201 44201 44201 ...  
## ..$ AQS_PARAMETER_DESC : Factor w/ 1 level "Ozone": 1 1 1 1 1 1 1 1 1 1 ...  
## ..$ CBSA_CODE : int [1:9737] 25860 25860 25860 25860 25860 25860 25860 25860 25860 25860 ...  
## ..$ CBSA_NAME : Factor w/ 17 levels "", "Asheville, NC",...: 9 9 9 9 9 9 9 9 9 9 ...  
## ..$ STATE_CODE : int [1:9737] 37 37 37 37 37 37 37 37 37 37 ...  
## ..$ STATE : Factor w/ 1 level "North Carolina": 1 1 1 1 1 1 1 1 1 1 ...  
## ..$ COUNTY_CODE : int [1:9737] 3 3 3 3 3 3 3 3 3 3 ...
```

```

## ..$ COUNTY : Factor w/ 32 levels "Alexander","Avery",...: 1 1 1 1 1 1 ...
## ..$ SITE_LATITUDE : num [1:9737] 35.9 35.9 35.9 35.9 35.9 ...
## ..$ SITE_LONGITUDE : num [1:9737] -81.2 -81.2 -81.2 -81.2 -81.2 ...
## $ : 'data.frame': 10592 obs. of 20 variables:
## ..$ Date : Factor w/ 365 levels "01/01/2019","01/02/2019",...: 1 2 3 ...
## ..$ Source : Factor w/ 2 levels "AirNow","AQS": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ Site.ID : int [1:10592] 370030005 370030005 370030005 370030005 370030005 ...
## ..$ POC : int [1:10592] 1 1 1 1 1 1 1 1 1 1 ...
## ..$ Daily.Max.8.hour.Ozone.Concentration: num [1:10592] 0.029 0.018 0.016 0.022 0.037 0.037 0.029 ...
## ..$ UNITS : Factor w/ 1 level "ppm": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ DAILY_AQI_VALUE : int [1:10592] 27 17 15 20 34 34 27 35 35 28 ...
## ..$ Site.Name : Factor w/ 38 levels "", "Beaufort",...: 33 33 33 33 33 33 ...
## ..$ DAILY_OBS_COUNT : int [1:10592] 24 24 24 24 24 24 24 24 24 24 ...
## ..$ PERCENT_COMPLETE : num [1:10592] 100 100 100 100 100 100 100 100 100 100 ...
## ..$ AQS_PARAMETER_CODE : int [1:10592] 44201 44201 44201 44201 44201 44201 44201 44201 ...
## ..$ AQS_PARAMETER_DESC : Factor w/ 1 level "Ozone": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ CBSA_CODE : int [1:10592] 25860 25860 25860 25860 25860 25860 25860 25860 ...
## ..$ CBSA_NAME : Factor w/ 15 levels "", "Asheville, NC",...: 8 8 8 8 8 8 8 8 ...
## ..$ STATE_CODE : int [1:10592] 37 37 37 37 37 37 37 37 37 37 ...
## ..$ STATE : Factor w/ 1 level "North Carolina": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ COUNTY_CODE : int [1:10592] 3 3 3 3 3 3 3 3 3 3 ...
## ..$ COUNTY : Factor w/ 30 levels "Alexander","Avery",...: 1 1 1 1 1 1 1 1 ...
## ..$ SITE_LATITUDE : num [1:10592] 35.9 35.9 35.9 35.9 35.9 ...
## ..$ SITE_LONGITUDE : num [1:10592] -81.2 -81.2 -81.2 -81.2 -81.2 ...
## $ : 'data.frame': 8983 obs. of 20 variables:
## ..$ Date : Factor w/ 365 levels "01/01/2018","01/02/2018",...: 2 5 8 11 14 ...
## ..$ Source : Factor w/ 1 level "AQS": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ Site.ID : int [1:8983] 370110002 370110002 370110002 370110002 370110002 ...
## ..$ POC : int [1:8983] 1 1 1 1 1 1 1 1 1 1 ...
## ..$ Daily.Mean.PM2.5.Concentration: num [1:8983] 2.9 3.7 5.3 0.8 2.5 4.5 1.8 2.5 4.2 1.7 ...
## ..$ UNITS : Factor w/ 1 level "ug/m3 LC": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ DAILY_AQI_VALUE : int [1:8983] 12 15 22 3 10 19 8 10 18 7 ...
## ..$ Site.Name : Factor w/ 25 levels "", "Blackstone",...: 15 15 15 15 15 15 15 15 ...
## ..$ DAILY_OBS_COUNT : int [1:8983] 1 1 1 1 1 1 1 1 1 1 ...
## ..$ PERCENT_COMPLETE : num [1:8983] 100 100 100 100 100 100 100 100 100 100 ...
## ..$ AQS_PARAMETER_CODE : int [1:8983] 88502 88502 88502 88502 88502 88502 88502 88502 ...
## ..$ AQS_PARAMETER_DESC : Factor w/ 2 levels "Acceptable PM2.5 AQI & Speciation Mass",...
## ..$ CBSA_CODE : int [1:8983] NA NA NA NA NA NA NA NA NA NA ...
## ..$ CBSA_NAME : Factor w/ 14 levels "", "Asheville, NC",...: 1 1 1 1 1 1 1 1 1 1 ...
## ..$ STATE_CODE : int [1:8983] 37 37 37 37 37 37 37 37 37 37 ...
## ..$ STATE : Factor w/ 1 level "North Carolina": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ COUNTY_CODE : int [1:8983] 11 11 11 11 11 11 11 11 11 11 ...
## ..$ COUNTY : Factor w/ 21 levels "Avery","Buncombe",...: 1 1 1 1 1 1 1 1 1 1 ...
## ..$ SITE_LATITUDE : num [1:8983] 36 36 36 36 36 ...
## ..$ SITE_LONGITUDE : num [1:8983] -81.9 -81.9 -81.9 -81.9 -81.9 ...
## $ : 'data.frame': 8581 obs. of 7 variables:
## ..$ Date : Factor w/ 365 levels "2019-01-01","2019-01-02",...: 3 6 9 12 15 18 21 24 27 ...
## ..$ Daily_AQI_Value : int [1:8581] 7 4 5 26 11 5 6 6 15 7 ...
## ..$ Site_Name : Factor w/ 25 levels "", "Board Of Ed. Bldg.",...: 14 14 14 14 14 14 14 14 ...
## ..$ AQS_Parameter_Desc: Factor w/ 2 levels "Acceptable PM2.5 AQI & Speciation Mass",...: 1 1 1 1 1 1 ...
## ..$ County : Factor w/ 21 levels "Avery","Buncombe",...: 1 1 1 1 1 1 1 1 1 1 ...
## ..$ Site_Latitude : num [1:8581] 36 36 36 36 36 ...
## ..$ Site_Longitude : num [1:8581] -81.9 -81.9 -81.9 -81.9 -81.9 ...

```

```
#dim(EPAAirList) this returns null when used with list. only works with single objects?
```

```
#2
```

TIP: All four datasets should have the same number of columns but unique record counts (rows). Do your datasets follow this pattern?

Wrangle individual datasets to create processed files.

3. Change any date columns to be date objects.
4. Create new dataframes with just the following columns:
'Date', 'DAILY_AQI_VALUE', 'Site.Name', 'AQS_PARAMETER_DESC', 'COUNTY', 'SITE_LATITUDE', 'SITE_LONGITUDE'
5. For the PM2.5 datasets, fill all cells in AQS_PARAMETER_DESC with "PM2.5" (all cell values in this column should be identical).
6. Save all four processed datasets in the Processed folder. Use the same file names as the raw files but replace "raw" with "processed".

```
#3
```

```
EPAAirOne$Date <- mdy(EPAAirOne$Date)
```

```
EPAAirTwo$Date <-mdy (EPAAirTwo$Date)
```

```
EPAAirThree$Date <- mdy(EPAAirThree$Date)
```

```
EPAAirFour$Date <-ymd(EPAAirFour$Date)
```

```
#4
```

```
EPAOne_DataFrame <- data.frame(  
  Date = EPAAirOne$Date,  
  Daily_AQI_Value= EPAAirOne$DAILY_AQI_VALUE,  
  Site_Name = EPAAirOne$Site.Name,  
  AQS_Parameter_Desc = EPAAirOne$AQS_PARAMETER_DESC,  
  County = EPAAirOne$COUNTY,  
  Site_Latitude = EPAAirOne$SITE_LATITUDE,  
  Site_Longitude = EPAAirOne$SITE_LONGITUDE,  
  stringsAsFactors = TRUE  
)
```

```
EPATwo_DataFrame <- data.frame(  
  Date = EPAAirTwo$Date,  
  Daily_AQI_Value= EPAAirTwo$DAILY_AQI_VALUE,  
  Site_Name = EPAAirTwo$Site.Name,  
  AQS_Parameter_Desc = EPAAirTwo$AQS_PARAMETER_DESC,  
  County = EPAAirTwo$COUNTY,  
  Site_Latitude = EPAAirTwo$SITE_LATITUDE,  
  Site_Longitude = EPAAirTwo$SITE_LONGITUDE,  
  stringsAsFactors = TRUE
```

```
)

EPAThree_DataFrame <- data.frame(
  Date = EPAAirThree$Date,
  Daily_AQI_Value= EPAAirThree$DAILY_AQI_VALUE,
  Site_Name = EPAAirThree$Site.Name,
  AQS_Parameter_Desc = EPAAirThree$AQS_PARAMETER_DESC,
  County = EPAAirThree$COUNTY,
  Site_Latitude = EPAAirThree$SITE_LATITUDE,
  Site_Longitude = EPAAirThree$SITE_LONGITUDE,
  stringsAsFactors = TRUE
)

EPAFour_DataFrame <- EPAAirFour #I think I somehow overrode the raw file with the processed one and I
```

#5

```
EPAThreee_DataFrame <- EPAThree_DataFrame %>%
  mutate(AQS_Parameter_Desc = "PM2.5")

EPAFourr_DataFrame <- EPAFour_DataFrame%>%
  mutate(AQS_Parameter_Desc = "PM2.5")
```

#6

```
#Saving to processed csvs

write.csv(
  EPAOne_DataFrame,
  row.names = FALSE,
  file=here('./Data/Processed/EPAair_03_NC2018_processed.csv')
)

write.csv(
  EPATwo_DataFrame,
  row.names = FALSE,
  file=here('./Data/processed/EPAair_03_NC2019_processed.csv')
)

write.csv(
  EPAThreee_DataFrame,
  row.names= FALSE,
  file=here('./Data/Processed/EPAair_PM25_NC2018_processed.csv')
)

write.csv(
  EPAFourr_DataFrame,
  row.names = FALSE,
  file=here('./Data/Processed/EPAair_PM25_NC2019_processed.csv')
)
```

Combine datasets

7. Combine the four datasets with `rbind`. Make sure your column names are identical prior to running this code.

```
Master_EPA_Air <- rbind(EPAOne_DataFrame, EPATwo_DataFrame, EPAThreee_DataFrame, EPAFourr_DataFrame)
```

8. Use code to display the dimensions of the combined dataset.

```
dim(Master_EPA_Air)
```

```
## [1] 37893      7
```

9. Wrangle your new dataset with a pipe function (`%>%`) so that it fills the following conditions:

- Include only sites that the four data frames have in common:
“Linville Falls”, “Durham Armory”, “Leggett”, “Hattie Avenue”,
“Clemmons Middle”, “Mendenhall School”, “Frying Pan Mountain”, “West Johnston Co.”, “Garinger High School”, “Castle Hayne”, “Pitt Agri. Center”, “Bryson City”, “Millbrook School”
- Some sites have multiple measurements per day. Use the split-apply-combine strategy to generate daily means: group by date, site name, AQS parameter, and county. Take the mean of the AQI value, latitude, and longitude.
- Add new columns for “Month” and “Year” by parsing your “Date” column (hint: `lubridate` package)
- Hint: the dimensions of this dataset should be 14,752 x 9.

10. Spread your datasets such that AQI values for ozone and PM2.5 are in separate columns. Each location on a specific date should now occupy only one row.

11. Call up the dimensions of your new tidy dataset.

12. Save your processed dataset with the following file name: “EPAair_O3_PM25_NC1819_Processed.csv”

```
#7 see above
```

```
#8 see above
```

```
#9
```

```
EPA_Common_Sites <- Master_EPA_Air %>%  
filter(Site_Name %in% c("Linville Falls", "Durham Armory", "Leggett", "Hattie Avenue",  
  "Clemmons Middle", "Mendenhall School", "Frying Pan Mountain",  
  "West Johnston Co.", "Garinger High School", "Castle Hayne",  
  "Pitt Agri. Center", "Bryson City", "Millbrook School")) %>%  
  mutate(month=month(Date), year=year(Date))
```

```
EPA_Summary <- EPA_Common_Sites %>% group_by(Date, Site_Name, AQS_Parameter_Desc, County, Site_Latitude, Site_Longitude)
```

```
## 'summarise()' has grouped output by 'Date', 'Site_Name', 'AQS_Parameter_Desc',  
## 'County', 'Site_Latitude', 'Site_Longitude', 'month'. You can override using  
## the '.groups' argument.
```

```

#IDK if I should use mutate here instead of summarize

#note to self...at this point the EPA_Summary dimensions are correct if I use summarise- still 14752 9,

EPA_Summary_AQS_Adjust <-EPA_Summary %>%
  pivot_wider(names_from = AQS_Parameter_Desc, values_from = Mean_AQI)

#This is not quite doing what I want so I think I need to surrender for now.

#11

dim(EPA_Summary_AQS_Adjust)

## [1] 8976    9

#dim are 8976 7....Sounds off.. Not sure where I went wrong

#12

write.csv(
  EPA_Summary_AQS_Adjust,
  row.names=FALSE,
  file=here('./Data/processed/EPAair_03_PM25_NC1819_Processed.csv')
)

```

Generate summary tables

13. Use the split-apply-combine strategy to generate a summary data frame. Data should be split into groups by site, month, and year. Then compute the mean AQI values for ozone and PM2.5 for each group. Finally, add a pipe to remove instances where the mean **ozone** values are not available (use the function **drop_na** in your pipe). It's ok to have missing mean PM2.5 values in this result.
14. Call up the dimensions of the summary dataset.

```

#13

EPA_Summaries <-
  EPA_Summary_AQS_Adjust %>%
  group_by(Site_Name,month,year,PM2.5, Ozone) %>%
  drop_na(Ozone)

#Question to John: Didn't we already computer the mean AQI in an earlier command? I'm probably misinter

#14

```

15. Why did we use the function **drop_na** rather than **na.omit**? Hint: replace **drop_na** with **na.omit** in part 13 and observe what happens with the dimensions of the summary date frame.

Answer: It looks like it deletes objects where the average AQS is missing/na - so the dimensions decrease accordingly.

16. Stage, commit, and push your Assignment to your GitHub account. Provide a link to your repository below.

Github repository URL: <https://github.com/>